

Program : Diploma in Wood and Paper Technology	
Course Code : 3128	Course Title: Workshop Practice
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience for the students with the basic wood workshop and machine shop to set firm foundation for higher semesters and life long learning.
- To familiarize basic manual and machining skills as an aid to function effectively in the process industry

Course Prerequisites:

Topic	Course code	Course Name	Semester
Basic Knowledge in Wood Working tools and Machine shop		Engineering Workshop practice	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Follow preliminary safety rules in workshop and select appropriate carpentry tool for the required application	6	Understanding
CO2	Demonstrate the handling of tools and familiarize the wood working machines	3	Applying
CO3	Construct wooden joints	15	Applying
CO4	Demonstrate machines with basic operations and their practice	18	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1							
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Follow preliminary safety rules in workshop and identify appropriate tools for the required application		
M1.01	Introduction of machine safety & floor safety precautions	2	Understanding
M1.02	Introduction of machine tools portable tools	2	Understanding
M1.03	Study of carpentry tools, carving tools & lathe tools	2	Understanding
CO2	Demonstrate the handling of tools and familiarize the wood working machines		
M2.01	Demonstrate the four side planer, finger joining machine, copy shaper, wide belt sander	1	Understanding
M2.02	Familiarization of wood working machines :band saw, resaw, circular saw ,surface planer, thickness planer, spindle moulder, chisel moulder, tenoning machine, chain mortise, panel saw, jig saw, multiple rip saw, radial arm saw, portable circular saw, router, mitre saw, wood working lathe ,air compressor with spray gun sanding machines(vibratory sander drum sander),drilling machine	2	Understanding
	Lab Exam - I	1	
CO3	Construct wooden joints		
M3.01	Construct lap joint, butt joint, mitred butt joint, rebate joint, tongue & grooved joint	15	Applying
CO4	Demonstrate machines with basic operations and their practice		
M4.01	Familiarisation of machines :lathe, drilling machine, boring machine, milling machine	3	Understanding

M4.02	Turning practices -Cylindrical turning, step turning, taper turning, form turning &thread cutting Drilling practices- Drilling machine identification, marking& drilling holes, boring &reaming Gear cutting practice Practice in Shaping Machine	15	Applying
	Lab Exam - II	2	

Text / Reference:

T/R	Book Title/Author
T1	Hajra Choudhary,A Text book of Workshop Technology,Media promoters and publishers
T2	R.S.khurmi,Workshop Technology,S Chand

Online Resources:

Sl. No	Website Link
1	https://youtu.be/CbKaIvFRMNI
2	https://youtu.be/n3fhq7YAzzY
3	https://youtu.be/liEWQAX5M4Y